σ-ADDITIVE PROBABILITY ON ORTHOMODULAR LATTICES: A SURVEY AND AN OPEN PROBLEM

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Abstract. On a Boolean algebra, a finitely coherent probability extends to a countably additive measure by classical machinery (Carathéodory; Loomis–Sikorski; Stone). On a non-distributive orthomodular lattice (OML) this machinery breaks. We survey the problem and isolate a precise open question, splitting it into two axes the Boolean case fuses: the *extension* axis (does a state extend to a charge on the dual?), a Horn–Tarski feasibility question settled negatively for $L(H)$; and the *descent* axis (does the charge concentrate on the physical points?). The combinatorial route fails — $\prod_n \mathrm{MO}_2$ is concrete, σ-complete and non-Boolean yet *segregated* (non-distributivity confined to finite blocks) — because a faithful set representation forces the OML Boolean (Theorem 5.2). Reading "no hidden realisation" as not being a mixture of dispersion-free states makes a relational measure a *contextual* state (Lemma 5.3). The open question: **does a non-Boolean, σ-complete, concrete OML carry a σ-additive contextual state whose contextuality is σ-essential** — witnessed by no finite sub-OML? The finite case is inhabited (Wright's pentagon), so only the σ-essential refinement remains, a compactness failure of CE type. Recent work on Dynkin-systems (Derr–Williamson [9]) settles it *negatively under a Polish/Borel-regularity hypothesis*, leaving the residue — whether a witness can be non-Polish-representable — a descriptive-set-theory question (Remark 5.1).

1. Introduction

All measurement is finite, yet the frameworks that organise empirical knowledge — probability theory, quantum mechanics — rest on infinite structures. The passage from finite observation to infinite structure is not automatic; it requires a commitment no finite body of evidence can compel. In probability theory the locus of that commitment is *countable additivity*:

$$A_1 \supseteq A_2 \supseteq \cdots, \quad \bigcap_n A_n = \varnothing \quad \Longrightarrow \quad \mu(A_n) \to 0.$$

De Finetti [8] held that only finite additivity is empirically grounded; Kolmogorov [26] adopted σ-additivity as an axiom.

The same tension recurs once observations need not be jointly performable. An observation *context* — a set of measurements performable together — is

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Boolean; gluing contexts along shared observations (a categorical colimit) returns:

$$\begin{aligned} \text{contexts mutually compatible} &\Rightarrow \text{Boolean algebra,} \\ \text{incompatible} &\Rightarrow \text{non-distributive OML.} \end{aligned}$$

(Gunji et al. [18]). The OML is *forced* by the context structure, not postulated; for non-quantum examples the incompatibility must be operationally imposed, since classical observables on a common space jointly distribute (Kolmogorov) and stay Boolean. The closed subspaces of a Hilbert space H form the prototype $L(H)$, not the premise.

On a non-distributive OML, Carathéodory's outer measure does not apply. For $L(H)$ it is rescued by Gleason's theorem [16], with no general-OML analogue known. Non-distributivity *per se* does not remove the two-valued homomorphisms the Boolean engine needs (MO_3 keeps them), but it removes the *guarantee*; for $L(H)$, $\dim \geq 3$, they fail outright (Kochen–Specker [25]). Whether enough survive is a question of *concreteness*, not non-distributivity: an OML is *concrete* (Definition 2.7) when it carries an order-determining set of two-valued states (Gudder [17]), and this is the qualifier on which the open problem turns.

The organising split. The question splits into two axes the Boolean case fuses: *extension* (does a state extend to a charge on $S_0(A)$? — settled, none does on $L(H)$) and *descent* (does the charge concentrate on the physical points? — open).

The relational standpoint. Following De Finetti [8] (probability without a presupposed sample space) and Birkhoff–von Neumann (propositions as a lattice, not subsets of a phase space), we start from the propositions and ask what their order alone determines — the measure-theoretic counterpart of pointless topology. The dual is then the McDonald–Bimbó space of *filters* (§2.3), and a positive answer would be a σ-additive probability theory on a non-distributive lattice, the non-Boolean analogue of localic measure theory (made precise in §5 as a *contextual* state; cf. [27]).

2. Preliminaries

We collect the lattice-theoretic apparatus; all notions are standard.

2.1. **Orthomodular lattices, and the orthogonal/meet-zero gap.**

Definition 2.1 (Orthocomplemented lattice)**.** A *bounded lattice* is a partially ordered set in which every pair a, b has a join $a \vee b$ and meet $a \wedge b$, with greatest element $\mathbf{1}$ and least element $\mathbf{0}$. An *orthocomplementation* is a map $a \mapsto a^\perp$ satisfying, for all a, b:

(i) $a \wedge a^\perp = \mathbf{0}$ and $a \vee a^\perp = \mathbf{1}$ (complement),
(ii) $a^{\perp\perp} = a$ (involution),
(iii) $a \leq b \Rightarrow b^\perp \leq a^\perp$ (order-reversing).

Definition 2.2 (Incomparable; meet-zero; orthogonal)**.** For elements a, b of an orthocomplemented lattice, three relations of increasing strength:

- *incomparable*: $a \not\leq b$ and $b \not\leq a$ — neither entails the other (an order condition);
- *meet-zero*: $a \wedge b = \mathbf{0}$ — no common refinement;
- *orthogonal*, written $a \perp b$: $a \leq b^\perp$ (equivalently $b \leq a^\perp$) — one entails the other's negation, *decisively incompatible*.

Each implies the one above it ($a \perp b \Rightarrow a \wedge b = \mathbf{0} \Rightarrow$ incomparable, for $a, b \notin \{\mathbf{0}, \mathbf{1}\}$). For two *distinct atoms*, incomparable and meet-zero coincide — an atom has only $\mathbf{0}$ below it, so $a \wedge b \in \{\mathbf{0}, a\}$, and $a \wedge b = a$ would force $a \leq b$; orthogonality remains strictly stronger.

Remark 2.3 (Meet-zero versus orthogonal)*.* The implication $a \perp b \Rightarrow a \wedge b = \mathbf{0}$ reverses in every Boolean algebra, but fails once the lattice is non-distributive: there meet-zero is strictly weaker than orthogonality. The distinctions developed below all turn on this separation.

Definition 2.4 (Orthomodular lattice)**.** An orthocomplemented lattice A is *orthomodular* if

$$a \leq b \quad \Longrightarrow \quad b = a \vee (a^\perp \wedge b). \tag{1}$$

A *Boolean algebra* is the special case where $\wedge$ distributes over $\vee$; every Boolean algebra is orthomodular, not conversely. The motivating non-Boolean example is $L(H)$, the closed subspaces of a Hilbert space with $a^\perp$ the orthogonal complement.

Definition 2.5 (Completeness notions)**.** Two countable-completeness conditions on an OML A are distinguished throughout, the second strictly weaker. Here $\bigvee_n a_n$ denotes the *join* (least upper bound) of $\{a_n\}$ in the order of A, unique when it exists; "$\exists \bigvee_n a_n \in A$" asserts that it does.

- A is *σ-complete* if
$$\forall \{a_n\}_{n\in\mathbb{N}} \subseteq A : \quad \exists \textstyle\bigvee_n a_n \in A$$
(equivalently every countable family has a meet, by $\bigwedge_n a_n = (\bigvee_n a_n^\perp)^\perp$);
- A is *σ-orthocomplete* if
$$\forall \{a_n\}_{n\in\mathbb{N}} \subseteq A \ \big(a_i \perp a_j \text{ for } i \neq j\big) : \quad \exists \textstyle\bigvee_n a_n \in A.$$

The orthogonal families are a subclass of all countable families, so σ-completeness quantifies over a strictly larger domain and is the stronger hypothesis: σ-completeness $\Rightarrow$ σ-orthocompleteness. Only the weaker condition is needed below. For a state s, σ-additivity is the requirement that

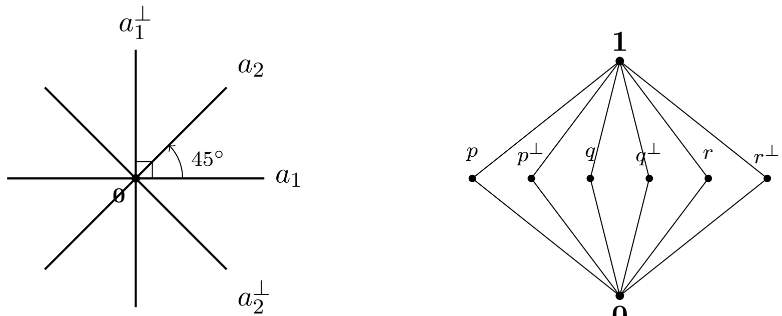
$$s\Big(\textstyle\bigvee_n a_n\Big) = \sum_n s(a_n)$$

for every countable pairwise-orthogonal family $\{a_n\}$. Its left side invokes $\bigvee_n a_n$ *only* for orthogonal families — so σ-orthocompleteness is exactly the hypothesis that makes it well-posed. Full σ-completeness is the stronger condition manipulated by the Loomis–Sikorski representation (§2.4).

Example 2.6 (MO$_n$, and the gap made concrete)**.** MO$_n$ is the lattice with $\mathbf{0}$, $\mathbf{1}$, and n complementary pairs of incomparable atoms, all sharing only $\mathbf{0}$ and $\mathbf{1}$. The geometric model is n lines through the origin: each line a is orthogonal to its perpendicular $a^{\perp}$, while two atoms from distinct pairs are non-perpendicular lines. It is orthomodular and non-distributive ($n \geq 2$): two atoms drawn from *distinct* complementary pairs have

$$a \wedge b = \mathbf{0} \quad \text{(meet only at the origin)} \qquad \text{yet} \qquad a \not\perp b,$$

so meet-zero $\neq$ orthogonal already here. MO$_3$ is the smallest non-distributive OML.



$a_1 \perp a_1^{\perp}$ (right angle),
$a_1 \not\perp a_2$ but $a_1 \wedge a_2 = \mathbf{0}$

complementary pairs $(p, p^{\perp}), (q, q^{\perp}), (r, r^{\perp})$;
any two non-paired atoms meet at $\mathbf{0}$, join at $\mathbf{1}$

FIGURE 1. MO$_n$ illustrated. *Left:* the geometric model for MO$_2$ — four atoms as lines through the origin in $\mathbb{R}^2$. The pair $a_1, a_1^{\perp}$ meets at a right angle (orthogonal), whereas a_1, a_2 meet at 45°: their lattice meet is still $\mathbf{0}$ (only the origin in common), yet they are *not* orthogonal — the meet-zero $\neq$ orthogonal gap. *Right:* the Hasse diagram of MO$_3$, the smallest non-distributive OML: three complementary pairs of incomparable atoms between $\mathbf{0}$ and $\mathbf{1}$.

Definition 2.7 (OMP; order-determining states; concrete logic)**.** An *orthomodular poset* (OMP) weakens Definition 2.4: joins $a \vee b$ are required only for *orthogonal* pairs. A set S of states on A is *order-determining* if it separates the order, i.e.

$$\big(s(a) \leq s(b) \text{ for all } s \in S\big) \implies a \leq b.$$

An OMP is a *concrete logic* (equivalently *set-representable*) if it embeds into $(\mathcal{P}(X), \subseteq, X \setminus (-))$ for some set X, preserving existing orthogonal joins — i.e. its propositions are modelled as actual *sets of outcomes*. Concreteness is equivalent to admitting an order-determining set of *two-valued* states (Gudder [17]), and it is the property that separates the open frontier from the settled cases (§5).

Remark 2.8 (Concreteness does not close the gap). A set-representable OMP needs only closure under complement and *disjoint* union [4]; intersection-closure would force a field of sets, hence Boolean. The lattice meet $a \wedge b$ is the greatest L-member inside $A \cap B$, so

$$\underbrace{A \cap B = \varnothing}_{\text{orthogonal}} \implies \underbrace{a \wedge b = \mathbf{0}}_{\text{meet-zero}}, \qquad \text{but not conversely.}$$

Concreteness does not force the converse: MO_3 is itself concrete (Definition 2.7; Gudder [17]), so the paradigmatic meet-zero-$\neq$-orthogonal example is already a concrete logic. What closes the gap is a richness/atomicity condition, not concreteness:

$$a \wedge b = \mathbf{0} \ \Rightarrow \ a \perp b,$$

equivalently: every nonempty $A \cap B$ contains a nonzero L-member — the abstract form being Tkadlec's *Boolean orthoposet* condition [36] (which does *not* mean Boolean *algebra*).

2.2. **The state/meet-additive/charge ladder.**

Definition 2.9 (State; meet-additive state; charge-extendible)**.** Let A be an OML. A *state* is a map $s : A \to [0,1]$ with $s(\mathbf{1}) = 1$ that is additive on *orthogonal* pairs: $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$; that is, a probability assignment whose additivity is guaranteed only on decisively incompatible propositions. Three strengthenings, differing in which pairs additivity is demanded on and strictly increasing in general (Example 2.10 on MO_3):

(1) *State*: $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$.
(2) *Meet-additive*: $s(a \vee b) = s(a) + s(b)$ whenever $a \wedge b = \mathbf{0}$ (binary, strictly stronger — it reaches the gap pairs of Definition 2.2). We avoid the word "valuation" here: in lattice theory it denotes the modular function $v(a)+v(b) = v(a \wedge b)+v(a \vee b)$, a different condition.
(3) *Charge-extendible*: $\sum_i s(a_i) \le 1$ for every pairwise-meet-zero family $\{a_i\}$ ($a_i \wedge a_j = \mathbf{0}$, $i \neq j$; strictly stronger again). This is the condition necessary for extension to a charge on the dual space (Definition 2.13) and the *extension* condition of §3.

Example 2.10 (The ladder is strict)**.** On MO_3 with complementary pairs $(p, p^\perp), (q, q^\perp), (r, r^\perp)$, the maximally mixed $s \equiv \frac{1}{2}$ is meet-additive (clears (2)) yet fails (3) at $\{p, q, r\}$: $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} > 1$. So no state on MO_3 is charge-extendible.

Definition 2.11 (Normal and singular states on $L(H)$)**.** On $A = L(H)$ a state s is

- *normal* if it is completely additive on every orthogonal family of projections,

$$s\Big(\bigvee_{i \in I} p_i\Big) = \sum_{i \in I} s(p_i) \quad \text{for all (not only countable) orthogonal } \{p_i\},$$

equivalently $s(\cdot) = \mathrm{tr}(\rho\,\cdot)$ for a density operator ρ (Gleason [16], $\dim \geq 3$);

- *singular* (purely) if

$$s(e) = 0 \quad \text{for every finite-rank projection } e.$$

Every state decomposes into a normal and a singular part (Takesaki); the two classes are exactly those distinguished by the arguments of §3.

2.3. **The McDonald–Bimbó dual space.**

Definition 2.12 (Dual space; representation map)**.** The McDonald–Bimbó duality [28] assigns to an OML A a compact space

$$S_0(A) = \big(F(A), \subseteq, \perp_A, P(A), T\big),$$

where $F(A)$ is the set of *filters* of A, ordered by $\subseteq$ and carrying an orthogonality relation $\perp_A$ and a Stone topology T, and $P(A) = \{\, \uparrow p : p \in A \,\} \subseteq F(A)$, with $\uparrow p = \{a \in A : a \geq p\}$, is the set of *principal* filters — the "physical points," the realisation datum (canonical, unlike the non-canonical pure points of the Boolean case). For $a \in A$ set $h(a) = \{x \in F(A) : a \in x\}$; then h is an OML isomorphism onto the $\perp$-stable clopens $\mathrm{CO}(S_0(A))^{\dagger}$. Define $\mu(h(a)) = s(a)$. Here $h(a)$ is the set of points at which a holds and μ is the state transported onto those sets; the open problem is whether μ is preserved under countable operations on them.

Definition 2.13 (Charge)**.** A *charge* on $S_0(A)$ is a map $\nu : \mathrm{CO}(S_0(A)) \to [0,1]$ on the *full* Boolean algebra of clopen subsets (not merely the $\perp$-stable ones in the image of h), with

$$\nu(S_0(A)) = 1, \qquad \nu(U \cup V) = \nu(U) + \nu(V) \ \text{ whenever } U \cap V = \varnothing.$$

A state s *extends to a charge* if some such ν satisfies $\nu(h(a)) = s(a)$ for all $a \in A$. Because h preserves meets, a pairwise-meet-zero family in A maps to disjoint clopens, so a charge necessarily satisfies $\sum_i s(a_i) \leq 1$ over such families — the charge-extendibility condition of Definition 2.9(3).

Remark 2.14 (The duality is finitary)*.* The representation map h satisfies

$$h\Big(\bigvee_{i\leq n} a_i\Big) = \Big(\bigcup_{i\leq n} h(a_i)\Big)^{\perp\perp} \quad \text{(finite)},$$
$$h\Big(\bigvee_{n} a_n\Big) \neq \Big(\bigcup_{n} h(a_n)\Big)^{\perp\perp} \quad \text{(countable)}:$$

finitary OML homomorphisms need not preserve infinite suprema. This is the central obstruction the open problem of §5 confronts, and it is self-dual: since $a \mapsto a^{\perp}$ is an order anti-isomorphism, $\bigwedge_n a_n = (\bigvee_n a_n^{\perp})^{\perp}$, so the failure for countable meets is the De Morgan dual of the failure for countable joins. The rival duality of Cannon–Döring [5] is also finitary and carries no measure; we use McDonald–Bimbó because it carries $P(A)$ as explicit structure.

2.4. **The Boolean engine: Stone duality.** In the Boolean case the descent equivalence

$$\sigma\text{-additive} \quad\Longleftrightarrow\quad \text{the measure concentrates on the physical points}$$

is proved through *Stone duality*, in four steps [27]:

(S1) a Boolean event algebra B embeds in the clopen algebra $\mathrm{CO}(\mathrm{St}\,B)$ of its Stone space;
(S2) a charge on B extends to a Baire measure on $\mathrm{St}\,B$ (Carathéodory);
(S3) σ-additivity $\Longleftrightarrow$ the measure vanishes on meagre Baire sets (Rao–Rao [34]);
(S4) meagre-null $\Longleftrightarrow$ the measure lives on the ultrafilters closed under countable intersection = the realised points.

Steps (S3)–(S4) are the measure-theoretic face of the *Loomis–Sikorski* representation,

$$\sigma\text{-complete Boolean algebra} \;\cong\; (\sigma\text{-tribe of sets})\,/\,(\sigma\text{-ideal}),$$

the countable-join bookkeeping that drives the equivalence. The contrast with the OML case is exact: the Boolean engine (Stone duality together with the Loomis–Sikorski representation) has *no* known OML analogue (§4.3). Supplying such an engine, or proving none exists, is the open problem of §5.

Remark 2.15 (Why the bridge breaks: meet-closed but not join-prime). A “realised” point x carries two conditions, which Rao–Rao's “vanishes on meagre sets” makes coincide in the Boolean case:

(M) closed under countable meets;
(J) $x \ni \bigvee_n a_n \Rightarrow x \ni a_n$ for some n (off the join-defect).

Non-distributivity separates them. For a principal filter $x = {\uparrow}p$, condition (M) always holds, whereas (J) can fail: $p \le \bigvee_n a_n$ does not force $p \le a_n$ for some n. In $L(H)$ this is witnessed by $a_n = \mathrm{span}(e_n)$ and $p = \mathrm{span}(e_1 + e_2)$:

$$p \;\le\; \bigvee\nolimits_n a_n = H, \qquad \text{yet} \qquad p \;\not\le\; a_n \;\; \text{for every } n.$$

So ${\uparrow}p$ is a realised point lying inside the join-defect $D = h(\bigvee_n a_n) \setminus \bigcup_n h(a_n)$. Topologically ${\uparrow}p$ is *isolated* in $S_0(L(H))$, hence:

$$D \text{ non-meagre (OML)} \qquad \text{vs.} \qquad D \text{ nowhere dense (Boolean).}$$

This is the finitary obstruction of Remark 2.14 on the dual: the category route (clopens modulo the meagre ideal) fails on the same D as the measure route, and reduces to the MacNeille completion of §4.3(ii).

3. Two axes: extension and descent

The central question — *when does a state s on an OML A extend to a σ-additive measure on $S_0(A)$?* — splits into two **independent** axes that the Boolean case fuses.

Extension.:
Does s extend to a finitely additive charge on the *full* Boolean clopen algebra of $S_0(A)$? This is governed by the meet-zero/orthogonal gap (§2.1) and is *settled*: it is a Horn–Tarski/Pitowsky feasibility question, a decidable linear program in the finite case, and can fail for every state.

Descent.:
Once extended, does the charge concentrate on the physical points $P(A)$? This is governed by σ-additivity — it needs the missing engine (§2.4) — and is *open*.

The extension axis is the abstract form of a familiar question: extending a finitely-additive assignment on incompatible propositions to a single joint distribution is exactly the feasibility problem behind the Bell inequalities (Fine [13]: a Bell model exists iff the correlations admit a joint distribution; Pitowsky [31]: the Bell inequalities are the facets of a correlation polytope), and Bell and Kochen–Specker non-classicality are jointly an extension problem for partial Boolean algebras (Budroni–Morchio [2], with the necessary-and-sufficient conditions of Horn–Tarski [22]). What is distinctive here is only that the extension is sought against the dual $S_0(A)$ rather than within a fixed scenario; the obstruction is the same finitary one, and it is settled. Two observations pin it down.

Proposition 3.1 (Finite case)**.** *For finite A the extension condition is a decidable linear program (Horn–Tarski* [22] */ Pitowsky correlation-polytope feasibility* [31]*; Budroni–Morchio* [2] *for the partial-Boolean-algebra formulation); since h preserves meets, meet-zero elements map to disjoint clopens and a charge must satisfy $\sum_i s(a_i) \le 1$ over pairwise-meet-zero families. It can fail for* every *state (*MO_3*, Example 2.10).*

Proposition 3.2 (Infinite $L(H)$, all states)**.** *Let* $\dim H = \infty$*.* No *state on $L(H)$ — normal or singular — extends to a charge on $S_0(L(H))$.*

Proof. The argument embeds a copy of MO_2 in $L(H)$ whose four atoms are infinite-dimensional, so that the obstruction reaches the singular states as well. Write $H = H_0 \otimes \mathbb{C}^2$ with H_0 infinite-dimensional, and fix an orthonormal basis e, f of $\mathbb{C}^2$. Set

$$a_1 = H_0 \otimes \mathbb{C}e, \quad a_1^{\perp} = H_0 \otimes \mathbb{C}f, \quad a_2 = H_0 \otimes \mathbb{C}(e+f), \quad a_2^{\perp} = H_0 \otimes \mathbb{C}(e-f).$$

These four infinite-dimensional subspaces form a copy of MO_2 in $L(H)$:

$$\begin{aligned} &a_1 \perp a_1^{\perp}, \quad a_2 \perp a_2^{\perp}, \quad a_i \vee a_i^{\perp} = H; \\ &a_i \wedge a_j = \mathbf{0} \ \text{ but } \ a_i \not\perp a_j \ \text{ (cross pairs).} \end{aligned}$$

Orthoadditivity with $s(\mathbf{1}) = 1$ gives $s(a_i) + s(a_i^{\perp}) = 1$ for $i = 1, 2$, using only additivity on orthogonal pairs, so it holds for *every* state. The family $\{a_1, a_1^{\perp}, a_2, a_2^{\perp}\}$ is pairwise meet-zero, and since h preserves meets ($h(a \wedge b) = h(a) \cap h(b)$), its members map to disjoint clopens; hence any charge satisfies

$\sum_i s(a_i) \le 1$. The two counts collide:

$$\sum_i s(a_i) = 2 \ \ \text{(orthoadditivity)} \qquad \text{versus} \qquad \sum_i s(a_i) \le 1 \ \ \text{(any charge)}.$$

The construction is finite ($n = 4$); σ-completeness of the lattice plays no role. □

The MO_2 count is standard (orthoadditivity together with $s(\mathbf{1}) = 1$; see [23]) and $\sum s(a_i) = 2$ is already state-independent in $L(\mathbb{C}^2)$. The role of the *infinite-dimensional* realisation is to extend the obstruction to the singular states ($s(e) = 0$ on every finite-rank e):

Atoms of MO_2	$s(a_i) + s(a_i^{\perp})$	**Reaches**
lines in $L(\mathbb{C}^2)$ (finite rank)	$= 1$ only if s charges finite rank	normal states only
$H_0 \otimes (\cdot)$, $\dim H_0 = \infty$	$= 1$ regardless of finite-rank behaviour	all states, incl. singular

Keeping each atom infinite-dimensional drops the finite-rank precondition of the line argument (Remark 3.3), and that is what closes the extension axis on $L(H)$ completely.

Remark 3.3 (An alternative argument for normal states only). A second, technique-disjoint argument reaches the normal states without the MO_2 family. If $s(e) > 0$ on some finite-dimensional e, it is positive on a 2-plane e_0; taking k pairwise-non-orthogonal lines $p_1, \dots, p_k \subset e_0$ yields $2k$ pairwise-meet-zero lines with $s(p_i) + s(p_i^{\perp}) = s(e_0)$, so a charge forces

$$k\, s(e_0) \le 1 \ \text{ for all } k \quad \Longrightarrow \quad s(e_0) = 0,$$

contradicting $s(e_0) > 0$. Lifting s to a functional via Mackey–Gleason / Bunce–Wright [3] ($B(H)$ has no type I_2 summand) and applying the Takesaki normal/singular decomposition identifies which states the argument reaches: it covers states with a nonzero normal part ($s(e) > 0$ for some finite-rank e), including every normal (Gleason) state, and misses the purely singular states ($s(e) = 0$ for all finite-rank e; e.g. ultrafilter vector states or Calkin-algebra pullbacks). Proposition 3.2 subsumes the missed case; this argument is recorded because its technique is independent and locates the singular states precisely.

4. What is known

The extension axis is settled — on $L(H)$ no state extends (Proposition 3.2); the descent axis turns on whether the Boolean engine of §2.4 has an OML analogue. Three things bear on that:

1. the Boolean benchmark such an analogue must match (§4.1);
2. the partial OML results, positive and obstructive (§4.2);
3. the routes already known to be blocked (§4.3).

4.1. **The Boolean benchmark.** The conditions an OML engine would have to reproduce. The Kelley–Vladimirov–Pták characterisation (Fremlin [14], Thm 391D) is complete: a Boolean algebra carries a strictly positive σ-additive measure iff it satisfies all three conditions below — each of which loses its meaning off the distributive world.

Boolean (KVP) condition	**OML analogue**
Dedekind σ-completeness	exists (σ-orthocompleteness)
weak (σ, ∞)-distributivity	none — relies on $\bigwedge$ distributing over $\bigvee$
chargeability	none — no structural characterisation on OMLs

4.2. **Positive and obstructive OML results. Positive** (require structure rich enough to approximate Hilbert-space geometry):

- Gleason [16] — $L(H)$, $\dim \geq 3$; σ-additivity *on the lattice.*
- Bunce–Wright [3] — JBW projection lattices; Chetcuti–Dvurečenskij [6].

Obstructive (Boolean machinery cannot be transplanted):

- Pták–Pulmannová [33] — a *finitary* condition (*subadditive*-unitality $s(a \vee b) \leq s(a) + s(b)$, no countable joins) already collapses the *lattice* to Boolean (their Thm 1); the OM*poset* version fails (Müller's set-representable counterexample [29]).
- Navara [30] — regularity does not force σ-additivity. Pták [32] — exotic state spaces via Greechie pasting.

4.3. **Three routes to a σ-OML engine, all unavailable.** The descent argument needs an OML analogue of the Boolean engine — a σ-Stone duality, or equivalently a Loomis–Sikorski representation. The three known routes are all unavailable, but not on the same footing: routes (i) and (ii) are closed by *theorem* (the hypothesis that would yield the representation provably forces Boolean, resp. leaves the category), whereas route (iii) is open in the strong sense — no construction is known, and no impossibility theorem rules one out (Remark 4.1). It is route (iii), restated, that is the Open Problem of §5.

(i) **RDP / effect-algebra.** Loomis–Sikorski holds for σ-MV algebras [11] and monotone σ-effect algebras *with* the Riesz Decomposition Property [12]; but a lattice effect algebra has RDP iff it is MV, and OML $\cap$ MV = Boolean (Kalmbach [23]), so the hypothesis that would yield the representation collapses the class:

$$\mathrm{OML} + \mathrm{RDP} \iff \mathrm{Boolean}.$$

The *sharp-skeleton* evasion — representing the OML as the *sharp* elements of an ambient RDP effect algebra, so that only the ambient carries RDP — also closes: in any RDP effect algebra the sharp elements coincide with the center and are hence Boolean (Jenča

2001, Cor. 4.3; via Greechie–Foulis–Pulmannová 1995), so a non-Boolean sharp skeleton witnesses RDP-failure. Route (i) is closed against this refinement too (σ-completeness is not on the load path). The MacNeille completion of an OML need not be orthomodular (Harding [19]); one cannot complete to absorb countable joins. The topological version of this route — clopens of $S_0(A)$ modulo the meagre ideal — is this same completion in another form (the regular-open algebra), and collapses non-distributivity (Remark 2.15).

(ii) **σ-Stone duality.** None is *known*, and none is *ruled out*: the existing dualities are finitary (McDonald–Bimbó, Remark 2.14) or equational with no set/tribe representation (Freytes [15], on σ-complete OMLs), but no theorem forbids a countable-join-preserving one. The one machinery that supplies a point-free σ-additive valuation — topos/Bohrification — *sidesteps* this route rather than settling it (its σ-additivity lives on a distributive object / per-context Boolean blocks; see §4.5), so route (iii) stays open but the known candidate-machinery is exhausted as a crack.

4.4. **Prior art on point-free quantum probability.** Every existing point-free or directed-context construction fails at least one of the two requirements — σ-additivity and a natively non-distributive structure — as tabulated below.

Gleason tradition: Varadarajan [37], Kalmbach [23], Pták–Pulmannová [33]: σ-additive on a non-distributive OML, but real-valued on a fixed lattice — not point-free.

Döring 2009: [10] point-free on the spectral presheaf over a directed poset — but provably only *finitely additive*, on a distributive Heyting algebra.

Bohrification (HLS): [20, 21] point-free and the internal valuation *is* σ-additive (Scott-continuous) — but factors through an internally *distributive* locale, σ-additive only per commutative context. Non-distributivity is tamed, not carried natively.

Localic valuations: Vickers [38], Simpson [35], Coquand–Spitters [7]: point-free σ-additive, but on *frames* — distributive by definition.

OML Stone dualities: McDonald–Bimbó [28] and Cannon–Döring [5]: purely structural, no measure, no σ-version.

Each entry satisfies some of the requirements but not all. The open case is their conjunction, satisfied by no existing construction:

$$\boxed{\text{point-free} \times \sigma\text{-additive} \times \text{natively non-distributive} \times \text{directed-system-built.}}$$

(The directed-context-yields-non-distributivity construction is not new — it is the shape of Bohrification and of colimit-over-Boolean-contexts generation. Novelty would lie only in the σ-additive $\times$ point-free $\times$ native combination.)

Remark 4.1 (The boundary of the negative results). Two qualifications delimit what is ruled out.

(i): **No impossibility theorem covers the concrete class.** The known no-go results are finite or two-valued, and the one positive occupant is non-concrete:

$$L(H)\ (\dim \geq 3): \quad \text{no two-valued states at all (Kochen–Specker [25]),}$$

a fortiori none order-determining. Concreteness is the qualifier on which the open frontier turns.

(ii): **No σ-Loomis–Sikorski exists**, and this is visible in the dualities: both take infinite joins as a closure, not a union,

$$\text{Cannon–Döring: } \mathrm{cls}(\textstyle\bigcup S_i); \qquad \text{McDonald–Bimbó: } (\textstyle\bigcup h(a_n))^{\perp\perp}.$$

This is not to be overstated: Gudder's concrete logics [17] *are* set-representable non-Boolean orthoposets — but as point-ful posets, not via a σ-Loomis–Sikorski theorem.

4.5. **What the topos route does and does not buy.** The topos / Bohrification programme is the one machinery supplying a point-free *and* σ-additive valuation on a quantum logic, so it is the natural candidate to crack route (iii). A mechanism-level read shows it is *the same wall, with tools*: the σ-additive valuation always lives on a distributive object, its countable additivity routed through per-context Boolean blocks. This is the topos analogue of the universal floor (§5.1).

Bohrification (Heunen–Landsman–Spitters). A state on A becomes a continuous probability valuation on the internal Gelfand spectrum $\Sigma(A)$, a compact regular *frame* [21, §2,§6], hence internally distributive; the continuity axiom (Def. 6.11) is localic σ-additivity on a frame, where $\bigvee$ distributes. The non-commutativity sits in the base poset $\mathcal{C}(A)$, not the measure. Theorem 6.19 bijects these valuations with measures on the non-distributive $\mathrm{Proj}(A)$ and so *looks* like a counterexample, but is not: by Definition 6.17(a) such a measure restricts to a σ-Boolean morphism on every countably complete Boolean sublattice — its countable additivity is defined block-by-block and glued by naturality, never crossing a non-commuting join. That is the antithesis of σ-essential contextuality. (And $\mathrm{Proj}(A)$ is the non-concrete Gleason / $L(H)$ object, Rmk 4.1, not the concrete class the frontier turns on.)

Döring–Isham (spectral presheaf). Measures live on $\mathrm{Sub}_{cl}(\Sigma)$, a Heyting algebra / locale — distributive, and *not* a σ-algebra [10]. The non-distributive $\mathrm{Proj}(H)$ enters only through daseinisation, a coarse-graining that, in Wolters'

words [39], "cannot preserve both [joins and meets], as [$\mathrm{Sub}_{cl}(\Sigma)$] is distributive whereas [$\mathrm{Proj}(H)$] is nondistributive." The measure is in general only *finitely* additive; σ-additivity is recovered only locally, for families orthogonal at a single context (one Boolean block) and only for normal states.
The f.a.-vs-σ diagnosis. The route lands on the programme's finitely-additive-versus-σ gap exactly: σ-additivity is available precisely where the object is distributive (the internal frame, or the per-context Boolean blocks where Loomis–Sikorski already applies), and degrades to finite additivity precisely when a single global measure is demanded across the non-distributive whole. Of the four requirements (point-free, σ-additive, natively non-distributive, directed-system-built) the topos route satisfies three and misses the crux: native non-distributivity. It *sidesteps* the wall — an internally distributive Gelfand spectrum is a design goal — rather than settling the σ-essential question, which stays open.

5. The open problem

Open Problem. Is there a non-Boolean, σ-complete, *concrete* OML carrying a σ-additive *contextual* state — a state with no global hidden joint distribution, equivalently one not lying in the closed convex hull of the lattice's dispersion-free (two-valued) states — whose contextuality is *σ-essential*, i.e. witnessed by no finite sub-OML? Or can one prove that no such state exists?

Remark 5.1 (Relation to Derr–Williamson 2023)*.* The problem above is **not plainly open**: it must be read against Derr–Williamson [9], which studies coherent probabilities on *pre-Dynkin-systems* (concrete orthomodular posets) and *Dynkin-systems* (concrete σ-complete OMLs) from the imprecise-probability side. Their Theorem 4.5 characterises finite-additive global extendability by a finite condition matching "non-contextual on every finite sub-OML," and their Theorem D.6 upgrades this to a *countably additive* extension to the ambient Borel σ-algebra *under the hypotheses* that the OML is realised as a Dynkin-system D_σ of Borel sets of a *Polish* space Ω with $\sigma(D_\sigma) = \mathrm{Borel}(\Omega)$ and inner-regular σ-blocks.

The link to the present formulation is one-directional but in the direction that matters. A point $\omega \in \Omega$ gives the evaluation δ_ω, which is a two-valued and (since $A_n \downarrow \varnothing \Rightarrow \omega \notin A_n$ eventually) σ-additive dispersion-free state; a countably additive ν on $\mathrm{Borel}(\Omega)$ with $\nu|_{D_\sigma} = w$ exhibits w as the barycentre $w(A) = \int \delta_\omega(A)\, d\nu$, hence $w \in \overline{\mathrm{conv}}(S^\sigma_{\mathrm{df}})$ — non-contextual. So σ-extendable $\Rightarrow$ non-contextual, and through Theorem D.6 the implication "finitely non-contextual everywhere $\Rightarrow$ globally non-contextual" holds *whenever the OML admits such a Polish Borel realisation.* **Under that hypothesis the σ-essential cell is empty: no witness exists.** (The converse, non-contextual $\Rightarrow$ σ-extendable, does *not* hold — the dispersion-free states properly contain the point evaluations δ_ω — but it is not needed for the negative conclusion.)

The residue is therefore a single condition, and it is precisely the representability the whole problem turns on: **can a σ-essential witness exist**

that admits *no* Polish Borel realisation (S^{σ}_{df} non-Polish / analytic-but-not-Borel)? This is a descriptive-set-theory question, and it is the same wall as the absence of a σ-Loomis–Sikorski representation (§4.3); the off-centre/non-band constraints of §6 are its lattice-side shadow. Beware the σ-version of the (A)/(B) point-space equivocation (Remark 5.6): Derr–Williamson's Polish hypothesis is on the underlying *point set* Ω, and obtaining such an Ω for an abstract concrete σ-complete OML is itself the missing engine, not a free input.

Here "relational" means *no hidden realisation*: the measure is not recoverable from an auxiliary space of definite states behind the propositions. Three point-spaces must be kept apart (the recurring trap of this subject): the *canonical* dual $P(A) = \{\uparrow p\}$ (one principal filter per element, §2.3); the *dispersion-free states* (the hidden-variable points); and the *external (A)-points* (Gleason's Hilbert rays, Remark 5.6). $P(A)$ is *permitted*; the objection is only to a measure built on a smuggled space — the (A)-points, equivalently a global hidden-variable joint over the dispersion-free states. By Lemma 5.3 a state has a hidden realisation iff it is a mixture of dispersion-free states, so a relational measure is precisely a *contextual* state. This is a condition on the dispersion-free states, not on $P(A)$: a witness is point-rich on $P(A)$ yet has no global dispersion-free joint.

5.1. **The reduction to contextuality, and what frames it.**
The universal floor. The naive route — realise the lattice as honest sets so that a measure becomes a measure on a space — is closed for every non-Boolean OML.

Theorem 5.2 (Universal floor)**.** *A* faithful set representation *of an OML L — an injective $h : L \to \mathcal{P}(\Omega)$ with $h(a \wedge b) = h(a) \cap h(b)$, $h(a \vee b) = h(a) \cup h(b)$, $h(a^{\perp}) = \Omega \setminus h(a)$ (a σ-tribe / Loomis–Sikorski representation when σ-joins are preserved) — exists only if L is Boolean.*

Proof sketch. Set operations $\cap, \cup, {}^c$ on $\mathcal{P}(\Omega)$ distribute, so the image $h(L)$ is a distributive sublattice; h injective and lattice-preserving makes L isomorphic to a distributive ortholattice, i.e. Boolean. □

Hence no relational measure can arise from a faithful descent to points. What a concrete OML *does* have (Definition 2.7) is an order-determining family of dispersion-free states; the live question is whether every state is a mixture of them.

The reduction. The following equivalence reduces "no hidden realisation" to a spanning condition on the dispersion-free states. Its content is not new — for a Bell scenario it is Fine's theorem [13] (a joint distribution exists iff a deterministic hidden-variable model does), and the convex-geometric form is Pitowsky's correlation-polytope picture [31]; the formulation for a general (concrete) OML, with dispersion-free states as the deterministic vertices, is the natural lift, in the lineage of the sheaf-theoretic account of contextuality [1].

Lemma 5.3 (Relational = contextual)**.** *Let L be a concrete OML with dispersion-free state space S_{df}, and w a state on L. Then w admits a hidden realisation — a probability measure on a space of global two-valued valuations of L inducing w — if and only if $w \in \overline{\mathrm{conv}}\,(S_{\mathrm{df}})$. We call w* contextual *when it is not.*

Proof sketch. A global two-valued valuation of L is exactly a dispersion-free state, so a hidden realisation is a probability measure μ on S_{df} with $w(a) = \int \omega(a)\, d\mu(\omega)$ for all a — i.e. w is the barycentre of μ. The barycentres of probability measures on the (weak-$*$ compact) set S_{df} are exactly $\overline{\mathrm{conv}}(S_{\mathrm{df}})$. The handle is *spanning*, not simplicity: non-uniqueness of μ is irrelevant; only its *existence* matters. □

The prize is therefore a concrete σ-complete OML with a σ-additive state outside the dispersion-free hull.

Why σ-essential. The finite version of this is already inhabited. Wright's pentagon [40] is a finite, concrete OML — a Greechie loop of five three-atom blocks, hence a lattice — with an order-determining family of dispersion-free states, yet it carries a non-classical $\{0, \frac{1}{2}\}$-valued state lying outside their closed convex hull. (The five intertwining atoms form a 5-cycle in which consecutive atoms share a block, so any dispersion-free state, taking at most one of each adjacent pair to 1, sets at most 2 of the 5 to 1; their hull has vertex-sum ≤ 2, whereas the $\{0, \frac{1}{2}\}$ state — a legitimate state, each block summing to 1 — has vertex-sum $\frac{5}{2}$. This is the KCBS inequality [24].) So "concrete OML with a contextual state" is a 1978 fact, and the general spanning statement is false. What Wright does *not* supply is contextuality that requires the countable structure: the pentagon's is witnessed by a finite sub-OML (itself). The prize must therefore be *σ-essential* — contextual in the σ-complete whole but in no finite sub-OML. This is not a device to evade Wright: "every finite piece has a global section, the countable whole has none" is a *compactness failure* of exactly the type by which countable additivity is not first-order axiomatizable over finitely additive coherence (see Remark 5.4).

Status. No witness has been constructed: the product $\prod_n \mathrm{MO}_2$ fails (its contextual states are only finitely additive — diffuse states on its central $P(\mathbb{N})$, which σ-additivity forces to concentrate on points, killing the contextuality, §5.2); Wright is finite; no other construction is known. On the negative side, the cell is now known to be *empty for every OML admitting a Polish Borel point-realisation* (Derr–Williamson, Remark 5.1). What remains genuinely open is therefore the *non-representable* case: a witness, if one exists, must admit no such realisation — which is precisely the absence of a σ-Loomis–Sikorski representation (§4.3), a descriptive-set-theory question.

Remark 5.4 (The compactness-failure analogy)*.* The σ-essential requirement places the open problem alongside a known Boolean phenomenon. Over a

Boolean algebra, countable additivity of a finitely additive charge is not first-order axiomatizable: an ultraproduct of σ-additive charges can be finitely additive but not σ-additive, so every finite (indeed first-order) test is passed while the global property fails [27]. A σ-essential contextual state would be the non-distributive analogue — finite sub-OMLs all classically interpretable, the obstruction carried only by the countable whole.

5.2. **What the combinatorial route settles, and why it is not enough.** It is tempting to look for the object among concrete σ-orthocomplete OMLs, on the grounds that a state's σ-additivity invokes $\bigvee_n a_n$ only for orthogonal families (Definition 2.5). This route is **settled, and it does not produce the object.** The property that would make σ-additivity genuinely exceed finite additivity on a concrete non-Boolean lattice is the interleaving condition

$$(\bigstar) \quad \exists\, p,\ \exists\, \{a_n\} \text{ infinite orthogonal}, \quad p \wedge a_n = 0\ (\forall n), \quad p \not\perp a_n\ (\forall n), \tag{2}$$

and $(\bigstar)$ is satisfied *trivially* by the plain product $\prod_n \mathrm{MO}_2$: take $a_n = a$ in coordinate n (zero elsewhere), and $p = \bigvee_n b_n$ where $b_n = b$ in coordinate n (zero elsewhere) — a genuine countable *orthogonal* join, so $p = (b, b, b, \dots)$ exists by σ-orthocompleteness. Then $p \wedge a_n = 0$ (coordinate n: $a \wedge b = 0$ in MO_2; elsewhere $a_n = 0$) and $p \not\perp a_n$ (coordinate n: $b \not\leq a^\perp$). Since $\prod_n \mathrm{MO}_2$ is concrete, σ-complete (a product of finite, hence complete, lattices), non-Boolean, and infinite, it satisfies $(\bigstar)$ and *every* combinatorial hypothesis one might impose — yet it is exactly the *segregated* object the open problem must exclude. So the combinatorial axioms (σ-completeness, concreteness, $(\bigstar)$) are *not* the discriminator; the missing ingredient is non-segregation, which is a point-free notion, not a closure property.

Remark 5.5 (The witness has no points, and the Boolean boundary). The universal closure above (a faithful set representation forces distributivity) already rules out the point-based route for every non-Boolean OML. The witness $\prod_n \mathrm{MO}_2$ shows the failure concretely and at the level of points: MO_2 admits *no* two-valued *homomorphism* — each of its four separating two-valued *states* ($a \mapsto 0$, $b \mapsto 0$, etc.) fails $\omega(a \vee b) \leq \omega(a) + \omega(b)$ since $a \vee b = \mathbf{1}$, so none preserves $\vee$ — and $\prod_n \mathrm{MO}_2$ inherits this, since its diagonal copy $\{\mathbf{0}, \mathbf{1}, (a, a, \dots), (a^\perp, \dots), (b, b, \dots), (b^\perp, \dots)\}$ is a sub-MO_2 on which any homomorphism would restrict to one of MO_2's (none exists). (This is special to MO_2 and its products — a generic non-Boolean OML such as $\mathbf{2} \times \mathrm{MO}_2$ *does* carry homomorphisms, e.g. the projection; the universal obstruction is the distributivity argument, not homomorphism-scarcity.) So the witness is concrete (Definition 2.7: order-determining two-valued *states*) and σ-complete yet has no points in the homomorphism sense at all — the point-free target is forced, not engineered. The same arithmetic records the sharp Boolean boundary: an OML is Boolean iff it admits a *unital* set of *subadditive* states — one with, for each nonzero a, a subadditive state giving a value 1 (Pták–Pulmannová [33], for OMLs). A concrete non-Boolean OML therefore cannot have its separating two-valued states form such a unital set:

some fail subadditivity (here all four of MO_2's fail $\omega(a \vee b) \leq \omega(a) + \omega(b)$). The forcing property to Boolean is thus subadditivity — a finitary condition — not σ-additivity.

A worked instance fixes the picture. Navara's construction [30] produces an infinite, σ-orthocomplete, rich OML whose inner block is a finite *stateless* Greechie logic — which is exactly what makes Navara's $\mathcal{L}$ non-concrete. Replacing the stateless block by MO_2 (write $\mathcal{L}_2$) removes the only source of non-concreteness: the assembly is a product of a sublogic of a Kalmbach horizontal sum (blocks glued only at $\{0,1\}$, *not* an atom-sharing Greechie loop), and pure horizontal sums of concrete logics are concrete, as are sublogics and products. So $\mathcal{L}_2$ is concrete, σ-orthocomplete, non-Boolean, and carries ($\bigstar$) — yet it is not a new object: it is $\prod_n \mathrm{MO}_2$ with extra scaffolding, segregated in the same way, and it likewise admits no σ-tribe representation. It is the cautionary example showing that exhibiting ($\bigstar$) and concreteness together is easy and does *not* reach the open problem.

The art constrains the open (σ-complete) problem from both sides without closing it: *no* impossibility theorem covers the concrete class (Remark 4.1), while the two *theorem-closed* routes to an engine — RDP and MacNeille (§4.3(i),(ii)) — are ruled out, leaving only the third, σ-Stone duality, which is itself open. What is required is not a new idea in the abstract but a representation in that gap: non-distributive enough to evade routes (i),(ii), yet σ-faithful where the known dualities are merely finitary.

Remark 5.6 (Why $L(H)$+Gleason does *not* already settle this — the (A)/(B) point-space equivocation). "Probability from relations, point-free" may *look* delivered by $L(H)$+Gleason. It is not: the appearance rests on conflating two point-spaces, and *realisation* is defined as concentration on the second. The two are (A) the Hilbert rays of H, which Gleason *requires* — the density ρ in $s = \mathrm{tr}(\rho\,\cdot)$ is built from them — and (B) the McDonald–Bimbó dual filters $P(A)$, which Gleason *removes*, but only by being a representation theorem. Gleason is thus the *most* point-ful result available (every lattice state corresponds to the point datum ρ). What $L(H)$ exhibits is a *separation*, not point-freeness: σ-additive measures exist on the lattice (Gleason, for normal states), yet on the dual $S_0(L(H))$ no state extends to a charge at all (Proposition 3.2), so a fortiori none concentrates on $P(A)$. It cannot serve as the existence proof required, so the point-free σ-additive structure of §5 is the only candidate engine.

5.3. **Two axes, two dictionaries — and the cell with no dictionary.** The recurring confusion of this subject (Remarks 5.5, 5.6) is best held as a 2×2 on two *independent* axes, which must not be collapsed into one.

- **Axis 1 — distributivity = commutativity.** One condition in two dialects: a family of projections commutes *iff* the lattice they generate is distributive. "Commutativity" is the operator word, "distributivity" the lattice word, for the *same* dividing line — not different sides.

- **Axis 2 — concreteness.** A non-Boolean lattice can be *point-rich in rays* (atoms = 1-dim subspaces) yet *point-free in valuations* (no two-valued homomorphisms, Kochen–Specker), or the reverse: the two senses of "point" come apart (the (A)/(B) equivocation, Rmk 5.6).

	distributive (commuting)	**non-distributive (non-commuting)**
concrete (valuations span)	classical: Boolean σ-algebra, Loomis–Sikorski free	the descent witness lives here: a concrete σ-complete non-Boolean OML ($\prod_n \mathrm{MO}_2$ lies here, but segregated)
non-concrete (KS: few valuations)	atomless measure algebra (valuation-poor)	$L(H)$: rays abundant, valuations KS-barred — the operator / e.s.a. world, the excluded pole

There is not one "lattice $\leftrightarrow$ space" dictionary but two, one per row: *Birkhoff–von Neumann* (bottom row) sends elements to closed subspaces and atoms to **rays**, with $a \vee b$ the closed span — rich in ray-points, KS-barred from valuation-points, hence non-concrete; *Stone / concrete-logic* (Gudder, top row) sends elements to sets of outcomes and atoms to points of Ω — rich in valuation-points, hence concrete. The operator question and the descent question share Axis 1 but sit in opposite rows: spectral projections *force* $L(H)$ (bottom-right), whereas the descent witness must be top-right. So the descent problem is the one cell no operator algebra can occupy — why a witness must be a pasting / countable colimit of concrete blocks, not a Hilbert-space construction, and why $L(H)$+Gleason cannot settle it. In this language the open problem (route (iii)) is exactly: *is there a logic $\leftrightarrow$ geometry dictionary for the top-right cell?* — and a σ-Loomis–Sikorski for concrete OMLs *is* that missing dictionary. (Orientation, not a new result; it guards against collapsing Axis 2 into Axis 1, i.e. treating "non-distributive" as "$L(H)$".)

6. Directions

With the extension axis closed on $L(H)$ (Proposition 3.2), the combinatorial route shown to deliver only segregated objects (§5.2), and the prize reduced to a σ-essential contextual state (§5.1), the problem has a clean dichotomy: two exits, each a genuine terminus. The negative exit is, moreover, *already taken in the Polish-representable case* (Derr–Williamson, Remark 5.1); both exits below therefore live or die on the same feature — whether a witness can fail to admit a regular Borel point-realisation.

Exhibit a σ-essential witness:

Construct an infinite concrete OML carrying a σ-additive contextual state that no finite sub-OML witnesses. The natural attack is a compactness-failure construction (cf. Remark 5.4): a concrete σ-complete OML whose finite sublogics are all classically interpretable (dispersion-free states span them) but whose countable joins force a Kochen–Specker-type obstruction only in the limit. The faithful set representation is ruled out at every stage (Theorem 5.2), so the obstruction must be carried by the σ-structure itself, not by any finite block. A necessary condition follows: the witness must place its infinitary structure *off-center* (irreducible, with non-central infinite blocks). A product of finite blocks such as $\prod_n \mathrm{MO}_2$ is excluded a priori, since it forces all infinitary content into the Boolean center, where σ-additivity concentrates the contextual states onto points and removes them (§5.2); a pasting or countable colimit of Wright-type blocks is the natural place to look.

Prove no such witness exists:

A spanning theorem: every σ-additive state on a σ-complete concrete OML lies in the closed convex hull of its dispersion-free states *except* where a finite sub-OML is already contextual (Wright). This would be an impossibility result — a σ-essential contextual state cannot exist, so relational σ-additive probability is operationally invisible (present finitely or not at all). *This is already established for Polish-representable OMLs* (Derr–Williamson, Remark 5.1); a full impossibility result must extend it to the non-representable case, where the Polish/Borel-regularity hypothesis of their Theorem D.6 is unavailable. The Boolean boundary — subadditivity of a unital set of states (Pták–Pulmannová, Remark 5.5) — is a natural lever; note that such an argument cannot reuse the $\prod_n \mathrm{MO}_2$ mechanism (concentration on the Boolean center), which is vacuous off-center.

A candidate direction now closed: the singular case for $L(H)$. One might have hoped a singular state would extend where no normal state does, providing the first explicit "extends but may fail to concentrate" example; Proposition 3.2 rules this out, catching every state, singular included. The reduction is recorded because the singular case was expected to require infinitary methods and does not (Remark 6.1).

Remark 6.1 (Finite versus countable: finite suffices). Is the obstruction to extending a singular state realised by a *finite* or only by a *countable* pairwise-meet-zero family? The two arguments divide on exactly this point. The line argument (Remark 3.3) is finitary but *vacuous* on a singular state ($s(e) = 0$ for every finite-rank e); the MO_2 argument with infinite-dimensional atoms

(Proposition 3.2) uses a finite family ($n = 4$) and gives $\sum_i s(a_i) = 2$ state-independently, catching the singular case as well. So the singular obstruction is *not* entangled with the descent-axis passage, as the vacuity of the line argument might suggest: it is realised by a finite family, hence a self-contained finitary problem, separable from the general construction and not governed by the finitary-to-σ passage.

Appendix A. Glossary of definitions

A self-contained list of the notions used in the paper. Entries that restate a numbered definition in the body cross-reference it; the remaining entries are stated here. Throughout, L is an orthomodular lattice with orthocomplementation $a \mapsto a^\perp$, bottom $\mathbf{0}$, top $\mathbf{1}$.

Lattice notions.

Orthocomplemented lattice; orthomodular lattice (OML).:
A bounded lattice with an order-reversing, involutive complement $a^\perp$ satisfying $a \vee a^\perp = \mathbf{1}$, $a \wedge a^\perp = \mathbf{0}$ (Definition 2.1); it is *orthomodular* if $a \leq b \Rightarrow b = a \vee (b \wedge a^\perp)$ (Definition 2.4). A *Boolean algebra* is a distributive OML.

Orthogonal; meet-zero; incomparable.:
$a \perp b$ means $a \leq b^\perp$; *meet-zero* means $a \wedge b = \mathbf{0}$. Orthogonal implies meet-zero; the converse fails off the Boolean case, and that gap drives the subject (Definition 2.2).

Compatible; block.:
a, b are *compatible* if they generate a Boolean subalgebra, equivalently $a = (a \wedge b) \vee (a \wedge b^\perp)$. A *block* is a maximal pairwise-compatible subset of L; it is a maximal Boolean sub-OML, and every OML is the union of its blocks. (In a concrete logic, compatibility of A, B is $A = (A \cap B) \cup (A \cap B^c)$.)

Completeness.:
L is *σ-complete* if every countable subset has a join; it is *σ-orthocomplete* if every countable *pairwise-orthogonal* family has a join (Definition 2.5).

Centre; irreducible.:
The *centre* $Z(L)$ is the set of elements compatible with all of L; it is a Boolean subalgebra. L is *irreducible* if $Z(L) = \{\mathbf{0}, \mathbf{1}\}$ (no nontrivial direct-product decomposition).

MO_n**;** $L(H)$**.:**
MO_n is the modular ortholattice with $\mathbf{0}, \mathbf{1}$ and n complementary atom-pairs (MO_2: atoms $a, a^\perp, b, b^\perp$, all distinct meets $\mathbf{0}$); it is non-distributive for $n \geq 2$. $L(H)$ is the lattice of closed subspaces of a Hilbert space H (orthocomplement = orthogonal complement, join = closed span), the prototype non-distributive OML.

Representations.

OMP; concrete logic; order-determining states.:
An *orthomodular poset* (OMP) weakens an OML to a partial join (only of orthogonal elements). L is a *concrete logic* (set-representable) if it embeds in some $\mathcal{P}(\Omega)$ with $a^\perp \mapsto \Omega \setminus a$ and orthogonal joins as disjoint unions, equivalently if it carries an *order-determining* (separating) family of two-valued states (Definition 2.7).

Faithful set representation; σ-tribe.:
An injection $h : L \to \mathcal{P}(\Omega)$ with $h(a \wedge b) = h(a) \cap h(b)$, $h(a \vee b) = h(a) \cup h(b)$, $h(a^\perp) = \Omega \setminus h(a)$ (preserving *all* meets and joins as $\cap, \cup$); a *σ-tribe* / Loomis–Sikorski representation also preserves countable joins. By Theorem 5.2 only Boolean L admits one. (Contrast a concrete logic, which preserves only *orthogonal* joins as unions.)

Dual space $S_0(A)$; canonical points $P(A)$.:
$S_0(A)$ is the McDonald–Bimbó space of filters of A; $P(A) = \{\uparrow p : p \in A\}$ is the set of principal filters, one per element (Definition 2.12).

States and measures.

State; meet-additive; charge.:
A *state* is $s : L \to [0, 1]$ with $s(\mathbf{1}) = 1$ and $s(a \vee b) = s(a) + s(b)$ for $a \perp b$ (Definition 2.9); a *charge* is a finitely additive set function on a Boolean algebra (Definition 2.13).

σ-additive state.:
A state s with $s\big(\bigvee_n a_n\big) = \sum_n s(a_n)$ for every countable pairwise-orthogonal $\{a_n\}$ (Definition 2.5).

Dispersion-free state.:
A two-valued state, i.e. a state taking only the values $0, 1$ (equivalently a $\{0, 1\}$-homomorphism on each block); S_{df} denotes their set and S^σ_{df} the σ-additive ones. These are the deterministic / hidden-variable points.

Normal; singular state on $L(H)$.:
s is *normal* if $s(\cdot) = \mathrm{tr}(\rho\, \cdot)$ for a density operator ρ; *singular* if s vanishes on every finite-rank projection (Definition 2.11).

Subadditive state.:
A state s with $s(a \vee b) \le s(a) + s(b)$ for *all* a, b (not only orthogonal pairs). A family of states is *unital* if for each $a \ne \mathbf{0}$ some member gives $s(a) = 1$.

The descent vocabulary.

Contextual state.:
A state w that is *not* in the closed convex hull $\overline{\mathrm{conv}}(S_{\mathrm{df}})$ of the dispersion-free states — equivalently (Lemma 5.3) a state with no hidden realisation, i.e. not a mixture of dispersion-free states. A non-contextual state is one that is such a mixture.

σ-essentially contextual state.:
A σ-additive state w on a σ-complete L that is *contextual* — $w \notin \overline{\mathrm{conv}}(S^\sigma_{\mathrm{df}})$

— yet whose restriction to *every finite sub-OML* of L is non-contextual. (All three clauses are required: σ-additive, globally contextual, finitely non-contextual. The finite version without the third clause is already inhabited, Wright's pentagon.)

Extension axis; descent axis.:
The *extension* axis asks whether a state extends to a charge on the full clopen algebra of $S_0(A)$ (a Horn–Tarski feasibility question); the *descent* axis asks whether such a charge concentrates on the physical points $P(A)$ (governed by σ-additivity). See §3.

Segregated.:
Of an OML: one whose non-distributivity is confined to finite blocks, so that $S_{\mathrm{df}} = S^{\sigma}_{\mathrm{df}}$ and no σ-essential leak is possible (e.g. $\prod_n \mathrm{MO}_2$, §5.2).

Prior-art objects (§5, Remark 5.1).

Pre-Dynkin-system; Dynkin-system.:
In Derr–Williamson's terminology, a *pre-Dynkin-system* on Ω is a concrete OMP (a family of subsets containing Ω, closed under complement and under disjoint unions); a *Dynkin-system* additionally is closed under countable disjoint unions — a concrete σ-complete OML.

σ-extendable.:
In Derr–Williamson, a countably additive probability μ_σ on a Dynkin-system $D_\sigma \subseteq \mathrm{Borel}(\Omega)$ (Ω Polish) is *σ-extendable* if it extends to a countably additive probability on $\mathrm{Borel}(\Omega)$.

Polish space; inner regular.:
A *Polish* space is a separable completely metrizable space. A measure μ is *inner regular* if $\mu(F) = \sup\{\mu(K) : K \subseteq F,\ K \text{ compact}\}$.

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